

VXLAN and EVPN: A Practical Overview

Virtual Extensible LAN (VXLAN) is an overlay encapsulation protocol that extends Layer 2 segments across a Layer 3 underlay network. Each segment is identified by a 24-bit VXLAN Network Identifier (VNI), which allows up to 16 million logical networks on a shared physical fabric.

Ethernet VPN (EVPN) provides the control plane for VXLAN. Instead of flood-and-learn, MAC and IP reachability information is distributed between VTEPs using Multiprotocol BGP. This reduces unnecessary flooding and enables active-active multihoming in leaf-spine designs.

Key Components

- Underlay: provides IP reachability between loopbacks (OSPF or eBGP)
- Overlay: the EVPN address family carries MAC and IP routes
- Data plane: UDP encapsulation with default destination port 4789

4789 24 16777216

Note: Verify the MTU on the underlay before migration. VXLAN adds 50 bytes of overhead, so an underlay MTU of at least 1550 is required.